

Loop Control: Managing Execution Flow

Sometimes we need finer control over our loops—to skip iterations or exit early based on certain conditions.

The `break` Statement: Exiting a Loop Early

The `break` statement immediately exits a loop when a certain condition is met:

```
1. prices = [105, 106, 104, 107, 103, 108, 99]
2. stop_loss = 100
3.
4. for price in prices:
5.     print(f"Price: {price}")
6.     if price < stop_loss:
7.         print("Stop loss triggered! Exiting position.")
8.         break
9.
```

In this example, we exit the loop when the price drops below our stop loss level, simulating how a trading algorithm might halt analysis once a critical threshold is breached.

The `continue` Statement: Skipping Iterations

The `continue` statement skips the current iteration and moves to the next one:

```
1. assets = ["AAPL", "BND", "VIX", "GOOGL", "SPY"]
2.
3. for asset in assets:
4.     # Skip volatility products - different analysis needed
5.     if asset == "VIX":
6.         print(f"Skipping {asset} - requires special analysis")
7.         continue
8.
9.     print(f"Performing standard analysis on {asset}")
10.
```